

PHY 6645 - Quantum Mechanics I - Fall 2011
Homework set # 8, due October 19

1. Let $H(\lambda)$ be the Hamiltonian of a system which depends explicitly upon a parameter λ . Consider the eigenvalues $E_n(\lambda)$ and eigenstates $|E_n(\lambda)\rangle$ of $H(\lambda)$. Show that

$$\langle E_n(\lambda) | \frac{dH}{d\lambda} | E_n(\lambda) \rangle = \frac{dE_n}{d\lambda} \quad , \quad (0.1)$$

for all n .

2. Show that the free particle one-dimensional Schrödinger equation is invariant with respect to Galilian transformations. Do this by showing that, when the transformation $x' = x - vt$, $t' = t$ is applied, the transformed wavefunction $\psi'(x', t') = f(x, t)\psi(x, t)$, with $f(x, t)$ a phase factor, is a solution with respect to the primed variables if $\psi(x, t)$ is a solution with respect to the unprimed ones. Find the phase factor $f(x, t)$, and show that the travelling wave solution $\psi(x, t) = Ae^{i(xt - \omega t)}$ transforms as expected.

3. Problems 7.3.1, 7.3.4 and 7.3.6 in Shankar's book.

1.

$$H(\lambda) |E_n(\lambda)\rangle = E_n(\lambda) |E_n(\lambda)\rangle$$

$$\frac{d}{d\lambda} H(\lambda) |E_n(\lambda)\rangle + H(\lambda) \frac{d}{d\lambda} |E_n(\lambda)\rangle$$

$$= \frac{dE_n}{d\lambda} |E_n(\lambda)\rangle + E_n(\lambda) \frac{d}{d\lambda} |E_n(\lambda)\rangle$$

$$\langle E_n(\lambda) | \frac{dH(\lambda)}{d\lambda} |E_n(\lambda)\rangle + \langle E_n(\lambda) | H(\lambda) \frac{d}{d\lambda} |E_n(\lambda)\rangle$$

$$= \frac{dE_n}{d\lambda} + E_n(\lambda) \langle E_n(\lambda) | \frac{d}{d\lambda} |E_n(\lambda)\rangle$$

$$\langle E_n(\lambda) | H(\lambda) = \langle E_n(\lambda) | E_n(\lambda)$$

$$\therefore \frac{dE_n}{d\lambda} = \langle E_n(\lambda) | \frac{dH}{d\lambda} |E_n(\lambda)\rangle$$

2.

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x, t) = +i\hbar \frac{\partial}{\partial t} \psi(x, t)$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x'^2} \psi'(x', t')$$

$$= -\frac{\hbar^2}{2m} \left(\underbrace{\frac{\partial x}{\partial x'}}_{=1} \frac{\partial}{\partial x} + \underbrace{\frac{\partial t}{\partial x'}}_{=0} \frac{\partial}{\partial t} \right)^2 f(x, t) \psi(x, t)$$

since $x = x' + vt'$ $t = t'$

$$= -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} f(x, t) \psi(x, t)$$

$$+i\hbar \frac{\partial}{\partial t} \psi'(x', t')$$

$$= i\hbar \left(\frac{\partial x}{\partial t'} \frac{\partial}{\partial x} + \frac{\partial t}{\partial t'} \frac{\partial}{\partial t} \right) f(x, t) \psi(x, t)$$

$$= i\hbar \left(v \frac{\partial}{\partial x} + \frac{\partial}{\partial t} \right) f(x, t) \psi(x, t)$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x'^2} \psi'(x', t') = i\hbar \frac{\partial}{\partial t'} \psi'(x', t')$$

for any $\psi(x, t)$ which solves the Schrödinger equation, provided

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2 f}{\partial x^2} \psi + 2 \frac{\partial f}{\partial x} \frac{\partial \psi}{\partial x} \right) = i\hbar \left[\psi \left(v \frac{\partial f}{\partial x} + \frac{\partial f}{\partial t} \right) + v f \frac{\partial \psi}{\partial x} \right]$$

for any such ψ . We must have:

$$\left\{ \begin{array}{l} -\frac{\hbar^2}{2m} \frac{\partial^2 f}{\partial x^2} = i\hbar \left(v \frac{\partial f}{\partial x} + \frac{\partial f}{\partial t} \right) \quad \text{and} \\ -\frac{\hbar^2}{m} \frac{\partial f}{\partial x} = i\hbar v f \end{array} \right. \quad \downarrow$$

$$f(x, t) = f(t) e^{-\frac{i}{\hbar} m v x}$$

$$i\hbar \frac{\partial f}{\partial t} = -v \hbar \frac{\partial f}{\partial x} + \frac{\hbar^2}{2m} \frac{\partial^2 f}{\partial x^2}$$

$$= \left[-v \left(-\frac{i}{\hbar} m v \right) + \frac{\hbar^2}{2m} \left(-\frac{i}{\hbar} m v \right)^2 \right] f$$

$$= \frac{1}{2\hbar} m v^2 f$$

$$f(x, t) = c e^{-\frac{i}{\hbar} m v x + \frac{i}{\hbar} \frac{1}{2} m v^2 t}$$

constant $c = 1$

since $f(x, t) = 1$ when $v = 0$

$$\psi'(x', t') = e^{-\frac{i}{\hbar} m v x + \frac{i}{\hbar} \frac{1}{2} m v^2 t} \psi(x, t)$$

v = boost velocity

Consider $\psi(x, t) = A e^{i(kx - \omega t)}$

In this state, the particle has

momentum $p = \hbar k$

energy $E = \hbar \omega$

ψ satisfies the Schrodinger equation

provided

$$E = \frac{1}{2m} p^2$$

The velocity of the particle is

$$v_p = \frac{1}{m} p$$

$$E = \frac{1}{2m} v_p^2$$

$$\Psi'(x', t') = A e^{+\frac{i}{\hbar} (m v_P - m v) x - \frac{i}{\hbar} (\frac{1}{2} m v_P^2 - \frac{1}{2} m v^2) t}$$

$$= A e^{\frac{i}{\hbar} m (v_P - v) (x' + v t') - \frac{i}{\hbar} (\frac{1}{2} m v_P^2 - \frac{1}{2} m v^2) t'}$$

$$= A e^{\frac{i}{\hbar} m (v_P - v) x' - \frac{i}{\hbar} \frac{m}{2} (v_P^2 + v^2 - v_P v) t'}$$

$$= A e^{\frac{i}{\hbar} m v_P x' - \frac{i}{\hbar} \frac{m}{2} v_P^2 t'}$$

with $v'_P = v_P - v$

S7.3.1

$$\psi'' + (2\epsilon - y^2) \psi = 0$$

$$\psi(y) = \sum_{n=0}^{\infty} A_n y^n$$

$$\sum_{n=0}^{\infty} A_n [n(n-1) y^{n-2} + 2\epsilon y^n - y^{n+2}] = 0$$

$$\sum_{n=0}^{\infty} y^n [A_{n+2} (n+2)(n+1) + 2\epsilon A_n - A_{n-2}] = 0$$

$$\text{with } A_{-1} = A_{-2} = 0$$

$$A_{n+2} = \frac{A_{n-2} - 2\epsilon A_n}{(n+2)(n+1)}$$

$$A_2 = \frac{-2\epsilon A_0}{2 \cdot 1}, \quad A_4 = \frac{A_0 - 2\epsilon A_2}{4 \cdot 3}$$

$$A_4 = A_0 \left(\frac{1}{4 \cdot 3} + \frac{4\epsilon^2}{4 \cdot 3 \cdot 2 \cdot 1} \right)$$

$$= A_0 \frac{1}{4 \cdot 3} \left(1 + \frac{(2\epsilon)^2}{2 \cdot 1} \right)$$

$$A_6 = \frac{1}{6.5} (A_2 - 2\epsilon A_4)$$

$$= \frac{A_0}{6.5} \left(\frac{-2\epsilon}{2.1} - 2\epsilon \left(\frac{1}{4.3} + \frac{4\epsilon^2}{4.3.2.1} \right) \right)$$

$$= \frac{A_0}{6.5} (-2\epsilon) \left(\frac{1}{2.1} + \frac{1}{4.3} + \frac{4\epsilon^2}{4.3.2.1} \right)$$

$$A_{2p} = \frac{1}{(2p)(2p-1)} (A_{2p-4} - 2\epsilon A_{2p-2})$$

$$= \frac{1}{(2p)(2p-1)} \left[\frac{1}{(2p-4)(2p-5)} (A_{2p-8} - 2\epsilon A_{2p-6}) - \frac{1}{(2p-2)(2p-3)} (A_{2p-6} - 2\epsilon A_{2p-4}) \right]$$

= ...

Indeed, the solution does not follow so readily.

S 7.3.4

$$\Delta = \left(\frac{\hbar}{m\omega} \right)^{1/2}$$

$$\langle n' | X | n \rangle = \int_{-\infty}^{+\infty} dx H_{n'}\left(\frac{x}{\Delta}\right) H_n\left(\frac{x}{\Delta}\right) \frac{x e^{-\left(\frac{x}{\Delta}\right)^2}}{\sqrt{\pi} \Delta (2^{n+n'} n! n'!)^{1/2}}$$

$$= \frac{\Delta}{\sqrt{2^{n+n'} \pi n! n'!}} \int_{-\infty}^{+\infty} dy H_{n'}(y) H_n(y) y e^{-y^2}$$

$$= \frac{\Delta}{\sqrt{2^{n+n'} \pi n! n'!}} \int_{-\infty}^{+\infty} dy \frac{1}{2} e^{y^2} H_{n'}(y) [H_{n+1}(y) + 2n H_{n-1}(y)]$$

$$= \frac{1}{2} \Delta \left[\delta_{n', n+1} \sqrt{2(n+1)} + 2n \delta_{n', n-1} \frac{1}{\sqrt{2n}} \right]$$

$$= \left(\frac{\hbar}{2m\omega} \right)^{1/2} \left[\delta_{n', n+1} \sqrt{n+1} + \delta_{n', n-1} \sqrt{n} \right]$$

$$\langle n' | P | n \rangle = \int_{-\infty}^{+\infty} dx \psi_{n'}(x) \frac{\hbar}{i} \frac{d}{dx} \psi_n(x)$$

$$= \frac{\hbar}{i} \frac{\Delta^{-1}}{\sqrt{2^{n+n'} \pi n! n'!}} \int_{-\infty}^{+\infty} dy H_{n'}(y) e^{-\frac{1}{2}y^2} \frac{d}{dy} H_n(y) e^{-\frac{1}{2}y^2}$$

$$= \frac{\hbar}{i} \frac{\Delta^{-1}}{\sqrt{2^{n+n'} \pi n! n'!}} \int_{-\infty}^{+\infty} dy H_{n'}(y) e^{-y^2} [H'_n(y) - y H_n(y)]$$

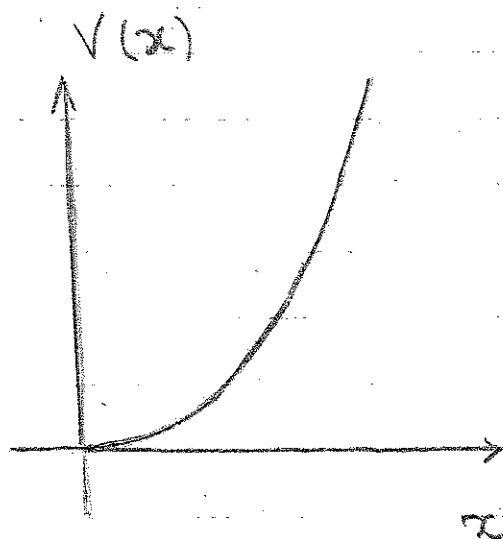
$$2n H_{n-1}(y) - \left(\frac{1}{2} H_{n+1}(y) + n H_{n-1}(y) \right)$$

$$= n H_{n-1}(y) - \frac{1}{2} H_{n+1}(y)$$

$$= \frac{\hbar}{i} \left[\delta_{n', n-1} n \frac{1}{\sqrt{2n}} - \frac{1}{2} \delta_{n', n+1} \sqrt{2(n+1)} \right] \Delta^{-1}$$

$$= \left(\frac{m \omega \hbar}{2} \right)^{1/2} \frac{1}{i} \left[\delta_{n', n-1} \sqrt{n} - \delta_{n', n+1} \sqrt{n+1} \right]$$

5.7.3.6



$$V(x) = \frac{1}{2} m \omega^2 x^2 \quad \text{for } x > 0$$

$$= \infty \quad \text{for } x \leq 0$$

$$\psi(x=0) = 0$$

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2 \right] \psi_n(x) = E_n \psi_n(x)$$

Clearly

$$E_n = \hbar \omega \left(n + \frac{1}{2} \right)$$

$$\psi_n(x) = \frac{1}{\sqrt{\sqrt{\pi} \Delta 2^n n!}} H_n\left(\frac{x}{\Delta}\right) e^{-\frac{1}{2} \frac{x^2}{\Delta^2}} \quad \text{for } x > 0$$

$$= 0$$

$$\text{for } x < 0$$

for n odd, are solutions.

There are no other solutions.

Indeed, let us assume there is another eigenfunction $\psi'(x)$ of eigenvalue E' .

$$\begin{aligned}\psi'(x) &= \psi_{E'}(x) && \text{for } x > 0 \\ &= 0 && \text{for } x < 0\end{aligned}$$

Consider the wavefunction

$$\begin{aligned}\psi''(x) &= \psi_{E'}(x) \frac{1}{\sqrt{2}} && \text{for } x > 0 \\ &= -\psi_{E'}(-x) \frac{1}{\sqrt{2}} && \text{for } x < 0\end{aligned}$$

$\psi''(x)$ is a eigenfunction of the harmonic oscillator hamiltonian with eigenvalue E' .

But then $\psi'(x)$ is not a new solution.